

pyladcp · SADCp processing with CODAS

raw VmDAS → edited + calibrated shipboard-ADCP product → pyladcp (--sadc-source codas) — one-page recipe

WHY CODAS

CODAS adds what raw .STA/.LTA averages lack: an editing pass (pflag), a watertrack calibration (misalignment angle + amplitude scale), and smoothed GPS navigation. On-station the raw constraint is already fine (40-cast MORIA A/B: interchangeable to ~1 cm/s) — CODAS matters UNDERWAY: phase ϕ biases $\approx \sin\phi \times \text{ship speed}$; an amplitude error of $a\%$ biases $\approx 10 \text{ cm/s per } \% \text{ at } 5 \text{ m/s}$. Use it for sections/transects and as an independent check of the raw chain.

ONE-TIME SETUP (~15 MIN)

```
bash scripts/codas_install.sh
# conda env pycodas + codas3/pycurrents/onship
# + demos + etopo (linked into the env)
```

PROCESS A CRUISE (MORIA OS150, ~5 MIN)

```
bash scripts/codas_moria_os150.sh
# other instruments: env overrides, e.g. OS75:
SADCp_DATA=../sadc_75/DATA SADCp_SONAR=os75nb \
SADCp_PROc=os75nb_sta SADCp_DB=aMORIA75 \
bash scripts/codas_moria_os150.sh
```

The script: (1) adcp_{tree}.py makes the tree; (2) stages NNN_₋prefixed CHRONOLOGICAL copies of the .STA files; (3) writes the control file; (4) runs quick_adcp.py --auto.

Never feed name-ordered files: parts of CODAS glob by FILENAME. On MORIA that misaligned every position by 682 ensembles and silently poisoned all absolute velocities — while the watertrack cal still looked fine. Always re-check the product's positions against a known cast before using it.

SINGLE-PING ROUTE (USE FOR SECTIONS / UNDERWAY)

```
bash scripts/codas_moria_os150_enr.sh
# OS75 (serial feeds swapped): ENR_GYRO_NR=2 + ENR_* overrides
```

STA averages are formed onboard from raw UNEDITED pings: underway, bins beyond the bubble/noise-limited range are garbage that keeps $pg \geq 50$ and survives average-level editing — MORIA deep (>300 m) underway rms u was 0.68 m/s (STA) vs 0.13 m/s after per-ping editing (on station both routes are clean). This route runs the full UHDAS chain on .ENR + N1R/N2R: per-ping editing BEFORE averaging, beam→earth with gyro + attitude correction, working bottom-track cal; it also lowered the apparent amplitude error +2.1% → +0.9%. Its workarounds (deployment-grouped staging, ping-number nav matching, topog link) are already encoded in the script.

CONTROL-FILE PARAMETERS THAT MATTER

--yearbase	year of first data; nc time = days since Jan 1
--datatype sta	release CODAS: sta/lta/uhdas/pingdata only
--sonar os150nb	instrument + ping type (nb = narrowband)
--ens_len 120	onboard STA interval [s] — read the .VMO
--max_search_depth	seabed search for editing; $\geq$ shelf depth

CALIBRATE — THE STEP --AUTO DOES NOT DO

```
cd <workdir>/os150nb_sta
tail -20 cal/watertrk/adcpcal.out # medians
quick_adcp.py --cntfile q_py.cnt --auto \
--steps2rerun rotate:navsteps:calib:matfiles:netcdf \
--rotate_angle 0.003 --rotate_amplitude 1.021
```

phase = transducer misalignment [deg]; amplitude = scale factor. Both errors grow with ship speed (invisible on station). Trust a cal only with a healthy point count, and verify the product afterwards.

Deliverable: contour/<sonar>.nc — absolute ocean u/v per ensemble (pflag editing applied), positions, uship/vship.
MORIA calcs: OS150 STA 1.021/0.003° · ENR 1.009/−0.09° · OS75 1.010/−0.22°.

USE IT IN PYLADCP (VMDAS STAYS THE DEFAULT)

```
ladcp-qa 80 --index .ladcp_archive.json \
--sadc <workdir>/os150nb_sta/contour/os150nb.nc \
--sadc-source codas
```

--sadc takes the .nc, its contour/ dir, or the processing dir; exports record provenance as sadc_{source}: codas:<path>. Keep vmdas for routine station work; pick codas for cruises with long steaming legs in cast windows or to use the edited series.

SECTION PLOTS

```
# raw VmDAS (GPS-leak screen --speed-max on by default)
ladcp-sadc-section --sadc ../sadc_150/DATA \
--by time -o sec_time.png
# CODAS product (already edited; cleaner underway)
ladcp-sadc-section \
--sadc <workdir>/os150nb_sta/contour/os150nb.nc \
--source codas --by distance \
--index .ladcp_archive.json -o sec_dist.png
```

--by time|distance picks the x-axis; --index draws LADCP station ticks; --max-depth/--clim crop and scale. --speed-max (median-speed blanking, 1.5 m/s) runs on either source: built for raw GPS leaks, also drops residual bad CODAS ensembles; 0 disables. --anomaly subtracts each ensemble's depth-mean → the baroclinic/shear view (barotropic signal removed, so weak structure shows at tight --clim).